# Putnam

Polynomials and their roots

Written by a model for Adit. Not Adit's words.

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## 1 Polynomials and their roots

A polynomial has three descriptions: its coefficients, its roots, and its values at enough points. Most polynomial problems are solved by moving from the description the problem gives to the one it hides.

*Recognise it.* A polynomial is given by a property of its roots ("all roots real", "roots in arithmetic progression") or of its values ("$p(k)$ is an integer for every integer $k$", "$p(1) = p(2) = 0$") rather than by its coefficients. An expression in several variables is symmetric, so the variables are the roots of something. A sequence of values is asked about at the next index. A question about the real roots of $p$ mentions $p'$, or could. A sum of coefficients taken every $k$-th one is asked for. A recurrence or congruence is really a statement about a polynomial identity.

### The model

Throughout, $p(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ has degree $n$, with coefficients in a field, which for the Putnam is $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{Z}/p\mathbb{Z}$.

**Theorem 1.1** (Factor theorem)**.** *$p(r) = 0$ if and only if $x - r$ divides $p(x)$.*

*Proof.* Divide $p$ by $x - r$ with remainder: $p(x) = (x - r)q(x) + c$ with $c$ a constant. Setting $x = r$ gives $c = p(r)$. □

**Theorem 1.2** (Degree counting)**.** *A nonzero polynomial of degree $n$ has at most $n$ roots, counted with multiplicity. Hence two polynomials of degree at most $n$ that agree at $n + 1$ points are identical, and a polynomial of degree at most $n$ with $n + 1$ roots is the zero polynomial.*

*Proof.* By the factor theorem, each root $r$ splits off a factor $x - r$ and lowers the degree by one, so after $n$ roots the quotient is a nonzero constant with no roots. For the second sentence, apply the first to the difference of the two polynomials. □

Over $\mathbb{C}$ every polynomial of degree $n$ has exactly $n$ roots with multiplicity; this is the fundamental theorem of algebra, used freely. So over $\mathbb{C}$, $p(x) = a_n \prod_{i=1}^{n} (x - r_i)$.

**Theorem 1.3** (Vieta)**.** *If $r_1, \dots, r_n$ are the roots of $p$ with multiplicity, then for each $k$,*

$$\sum_{i_1 < \cdots < i_k} r_{i_1} \cdots r_{i_k} = (-1)^k \frac{a_{n-k}}{a_n}.$$

*In particular $\sum r_i = -a_{n-1}/a_n$ and $\prod r_i = (-1)^n a_0/a_n$.*

*Proof.* Expand $a_n \prod (x - r_i)$ and compare the coefficient of $x^{n-k}$ with $a_{n-k}$. □

Vieta turns a symmetric expression in the roots into an expression in the coefficients. Power sums $\sum r_i^k$ are reached from the elementary symmetric functions either by Newton's identities or, faster on the day, by using $p(r_i) = 0$ to reduce $r_i^k$ to lower powers, as in the first example below.

**Theorem 1.4** (Rolle, for real roots)**.** *Let $p$ have real coefficients. Between two distinct real roots of $p$ there is a real root of $p'$. Consequently, if $p$ has $r$ real roots counted with multiplicity, then $p'$ has at least $r - 1$; equivalently, $p$ has at most one more real root than $p'$.*

*Proof.* The first sentence is Rolle's theorem from calculus. For the second, let the distinct real roots be $x_1 < \cdots < x_s$ with multiplicities $m_1, \dots, m_s$ summing to $r$. A root of multiplicity $m_i \geq 2$ of $p$ is a root of multiplicity $m_i - 1$ of $p'$, and each of the $s - 1$ open gaps contains a further root of $p'$. So $p'$ has at least $\sum (m_i - 1) + (s - 1) = r - 1$ real roots. □

Used the other way round: if $p'$ has few real roots, so does $p$. If $p'$ has $k$ real roots then $p$ has at most $k + 1$, and the sign of $p$ at the roots of $p'$ decides how many of those are achieved.

**Theorem 1.5** (Lagrange interpolation)**.** *Let $x_0, \dots, x_n$ be distinct and let $y_0, \dots, y_n$ be arbitrary. There is exactly one polynomial $p$ of degree at most $n$ with $p(x_i) = y_i$ for every $i$, namely*

$$p(x) = \sum_{i=0}^{n} y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

*Proof.* Each product is a polynomial of degree $n$ equal to $1$ at $x_i$ and $0$ at the other nodes, so the sum has the right values. Uniqueness is degree counting: two such polynomials differ by a polynomial of degree at most $n$ with $n + 1$ roots. □

**Theorem 1.6** (Finite differences)**.** *Define $\Delta p(x) = p(x + 1) - p(x)$. If $p$ has degree $n \geq 1$ with leading coefficient $a_n$, then $\Delta p$ has degree $n - 1$ with leading coefficient $n a_n$. Hence $\Delta^n p$ is the constant $n!\, a_n$, and $\Delta^{n+1} p = 0$, which written out at $x = 0$ reads*

$$\sum_{k=0}^{n+1} (-1)^{n+1-k} \binom{n+1}{k} p(k) = 0.$$

*Proof.* $(x + 1)^n - x^n = n x^{n-1} + \dots$, and lower terms of $p$ contribute lower degrees. The displayed identity is $\Delta^{n+1} p(0)$ expanded by the binomial theorem applied to the operator $\Delta = E - 1$, where $Ep(x) = p(x + 1)$. □

The identity gives $p(n + 1)$ from $p(0), \dots, p(n)$ without finding $p$. It is also the reason a polynomial that takes integer values at $0, 1, \dots, n$ takes integer values at every integer: the differences are integers, and $p(k)$ is a sum of binomial coefficients times differences.

**Theorem 1.7** (Roots-of-unity filter)**.** *Let $\omega = e^{2\pi i/m}$. For any polynomial $p(x) = \sum a_k x^k$ and any residue $r$ modulo $m$,*

$$\sum_{k \equiv r \pmod m} a_k = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr}\, p(\omega^j).$$

*Proof.* The right side is $\frac{1}{m}\sum_k a_k \sum_{j=0}^{m-1} \omega^{j(k-r)}$. The inner sum is $m$ when $m$ divides $k-r$ and otherwise is a geometric series with ratio $\omega^{k-r} \neq 1$ summing to $(\omega^{m(k-r)} - 1)/(\omega^{k-r} - 1) = 0$. □

With $m = 2$ this is the familiar $\sum a_{2k} = (p(1) + p(-1))/2$.

*First moves.* Write the polynomial by its roots, with the leading coefficient in front, not by its coefficients. Evaluate at $0$, $1$, $-1$, and if anything is cyclic, at the roots of unity. Take a derivative and apply Rolle. Reduce the coefficients modulo $2$. For a polynomial given by values, write down Lagrange or the finite-difference table for the small case and see what it says. For a symmetric expression in several variables, name the variables as the roots of a polynomial and let Vieta translate. Count degrees on both sides of any identity before believing it.

### Worked examples

**Example 1.8.** Let $r$, $s$, $t$ be the three roots of $x^3 - 2x - 5$. Compute $r^5 + s^5 + t^5$.

*Solution.* By Vieta, $r + s + t = 0$, $rs + st + tr = -2$, $rst = 5$. Then

$$r^2 + s^2 + t^2 = (r + s + t)^2 - 2(rs + st + tr) = 0 + 4 = 4.$$

Each root satisfies its own equation, $r^3 = 2r + 5$, so

$$r^5 = r^2 \cdot r^3 = r^2(2r + 5) = 2r^3 + 5r^2 = 2(2r + 5) + 5r^2 = 5r^2 + 4r + 10,$$

and the same for $s$ and $t$. Summing,

$$r^5 + s^5 + t^5 = 5 \cdot 4 + 4 \cdot 0 + 3 \cdot 10 = 50.$$

□

Nothing was computed about the roots themselves; the polynomial has one real root and two complex ones, and it does not matter. The move "reduce $r^k$ using $p(r) = 0$" is the polynomial version of reducing modulo something, and it works for any power sum.

**Example 1.9.** Show that for every real $a$ the polynomial $x^5 - 5x + a$ has at most three real roots, and determine the values of $a$ for which it has exactly three distinct real roots.

*Solution.* Let $p(x) = x^5 - 5x + a$. Then $p'(x) = 5x^4 - 5 = 5(x^2 - 1)(x^2 + 1)$, which has exactly two real roots, $x = -1$ and $x = 1$. By the Rolle count, $p$ has at most $2 + 1 = 3$ real roots with multiplicity.

For the second part, $p$ is increasing on $(-\infty, -1)$, decreasing on $(-1, 1)$, and increasing on $(1, \infty)$, with $p(x) \to -\infty$ as $x \to -\infty$ and $p(x) \to +\infty$ as $x \to +\infty$. Its local maximum is $p(-1) = -1 + 5 + a = a + 4$ and its local minimum is $p(1) = 1 - 5 + a = a - 4$. On each

of the three monotone pieces $p$ has at most one root, and it has a root on all three exactly when the local maximum is positive and the local minimum is negative, that is, when $a+4>0$ and $a-4<0$. So $p$ has exactly three distinct real roots if and only if $-4<a<4$. At $a=\pm 4$ one root is double, and for $|a|>4$ there is exactly one real root. □

The derivative had a simple factorisation and the problem fell to a sign table. Most "how many real roots" questions on the Putnam are this argument with a less convenient derivative, and the derivative's own real roots are found by the same method one level down.

**Example 1.10.** Let $p$ be a polynomial with real coefficients such that $p(x)=p(2-x)$ for all real $x$. Show that $p(x)$ is a polynomial in $(x-1)^2$.

*Solution.* Put $q(y)=p(1+y)$, a polynomial in $y$. The hypothesis at $x=1+y$ says $q(y)=p(1+y)=p(1-y)=q(-y)$ for all real $y$. Write $q(y)=\sum c_k y^k$. Then $q(y)-q(-y)=2\sum_{k\text{ odd}} c_k y^k$ is a polynomial vanishing at every real $y$, so by degree counting it is the zero polynomial and every odd coefficient $c_k$ is $0$. Hence $q(y)=\sum_j c_{2j}y^{2j}$ and $p(x)=q(x-1)=\sum_j c_{2j}(x-1)^{2j}$. □

A functional identity between polynomials becomes a statement about coefficients only through degree counting: "equal at infinitely many points" is what licenses "equal as polynomials". The shift to the symmetric point is the standard first move for any identity of the form $p(x)=p(c-x)$.

*Trap.* Roots that are complex or repeated. Vieta counts with multiplicity and over $\mathbb{C}$; "the roots are real" is a hypothesis to be used, not assumed. A polynomial identity checked at finitely many points without counting degrees proves nothing. The leading coefficient forgotten when writing $p$ by its roots: $p(x)=a_n\prod(x-r_i)$, and $a_n$ is often what the problem is about. Rolle gives an upper bound on real roots, never a lower one; the lower bound needs sign changes. Lagrange interpolation produces the unique polynomial of degree at most $n$; a polynomial of higher degree through the same points is not determined. The finite-difference identity needs equally spaced nodes.

**Ladder.**

1. 2019 A1. A symmetric cubic expression in three integers; which values it takes. Factor the symmetric form.
2. 2023 A2. An even-degree polynomial with prescribed values at prescribed points; find its other roots. Build $p$ from its values.
3. Roots #6. A polynomial with coefficients from a two-element set; can all its roots be real. Vieta against an inequality.
4. Roots #7. A polynomial derived from one with all roots real; show the derived one also has. Rolle and sign changes.
5. 2016 A1. Derivatives of an integer polynomial; the smallest order always divisible by a given integer. Polynomials meet congruences.
6. 2022 B1. A power series built from an integer polynomial; show its coefficients are all nonzero. Work modulo $2$.
7. 2024 A2. Iterates of a polynomial and a divisibility relation; which polynomials. Degree counting on the iterate.
8. 2017 A2. A rational recurrence whose terms are integer polynomials. The polynomial identity behind the recurrence.

9. 2019 B5. A polynomial that interpolates a famous recurrence at many points; its value at the next one. Finite differences.
10. 2018 B2. A polynomial with decreasing coefficients; no roots in a disc. A bound on the roots through a telescoping product.
11. 2019 A3. Polynomials with increasing coefficients; the smallest mean of the root moduli. Vieta and the AM–GM inequality.
12. 2022 A2. The square of a polynomial; how many coefficients can be negative. Extremal on the coefficient pattern.

Seminar: Roots #1, #8, #11a; the set notes that #11(b) is considerably harder, that #23 is the Chebyshev bound, and that #25 is Descartes's rule of signs.

*Where it hides.* Recurrences (2017 A2, 2019 B5): a sequence of polynomials, or a polynomial through a sequence's values, is a polynomial problem with a recurrence in front. Congruences (2016 A1, 2022 B1): an integer polynomial reduced modulo a prime is still a polynomial, with at most as many roots as its degree, which is the bound most such problems need. Inequalities: a symmetric inequality in $n$ variables is often Vieta applied to the polynomial with those roots. Linear algebra: the characteristic polynomial, and the space of polynomials of degree at most $n$ as a vector space of dimension $n+1$. Generating functions: a generating function is a polynomial or power series, and extracting every $k$-th coefficient is the roots-of-unity filter.